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CRITICAL CARE

Management of ROSC

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Key Points

- Early diagnosis is key. Refer all patients with cardiovascular instability or ECG evidence of ischaemia for coronary angiography, even if no ST elevation present.
- Consider CT head and CTPA prior to transfer to ICU.
- Do not intubate unless ongoing coma after ROSC. Do not sedate on ICU sedate unless RASS >0.
- Targeted Temperature Management (TTM) of 32-36 degrees for 24hours. Avoid hyperthermia (>37.7°C) for 72 hours.
- Prognostication is not possible for 72 hours, and should be lead by an ICM consultant.

Introduction

Post cardiac arrest care is a common presentation for critical care admission, both from in-hospital and out of hospital cardiac arrest. This guideline aims to guide anaesthetic and ICM staff to manage the initial investigation and support of the post-ROSC patient to optimise survival and rehabilitation.

Diagnosis/Admission Planning

1) Diagnosis: If there is clinical (e.g. haemodynamic instability) or ECG evidence of coronary ischaemia, refer for urgent Angiography first. **PPCI co-ordinator 01745 534142**. Then consider CT head or CTPA if coronary angiography is negative.
2) If no evidence of ischaemia, consider CTPA and/or CT head, especially if preceded by SOB/hypoxemia or headache/seizures/neurological deficit.
3) Contact ICU consultant on-call (Blp. 103) to discuss diagnosis, admission and specifics of care.
4) Contact ICU Nurse in charge to arrange bed preparation and agree admission timeline.

Clinical Care

A/B: Target: pO2 10-13kPa, pCO2 4.5-6kPA, LPV (6ml/kg IBW, PEEP ≥5, PAW <30)

If patient remains comatose or needs intubation for other reasons, perform RSI as per DAS critical care Intubation guideline. Otherwise if targets not met, consider NHFO or CPAP/NIV. Use temperature corrected values, including if TTM instituted.

C: Target: MAP >65mmHg, Lactate Falling.

Noradrenaline 1st line for hypotension. Consider dobutamine if lactate fails to improve or clinical evidence of hypoperfusion. Avoid metaraminol except as holding measure during CVC insertion.

If Fluids inotropes and pressors are inadequate to maintain MAP and improve lactate, discuss patient with consultant regarding ECMO referral.

Echo (FICE/CUSIC ± departmental echo) ASAP for all ROSC patients to determine cardiac pathology and degree of myocardial dysfunction.

Procedures: Arterial Line, CVC if needing pressors or inotropes. Consider IO if urgently needs central access.

D: If comatose, institute 'Targeted Temperature Management' for 24hrs using arctic sun, irrespective of location of arrest or rhythm. Target temperature range 32-36°C, 36°C optimal. If temperature not being met, consider atracurium infusion.

Do not sedate unless under influence of neuro-muscular blockers, RASS greater than '0' or requiring sedation to support treatment of B or C. Use short acting agents. Avoid thiopentone and benzodiazepines.

Avoid hyperthermia >37.7°C for >72hrs.

Maintain glucose 4-10mmol using VRII if more than one raised value.

Treat seizures with Levetiracetam 30mg/Kg. do not give anti-epileptics prophylactically.

Renal: Target UOP >0.5ml/kg/Hr. if not met, consider raising MAP to 70mmHg. Do not use higher target MAP without consultant approval.

Target K+ 4-4.5mmol/L, Mg++ >1.0mmol/L, Ionised Ca++ >1.0mmol/L, Phosphate >0.7mmol/L

GI: Commence trophic feeding at 10ml/hr if TTM instituted, otherwise, full feed once lactate improving.

Other: DVT and Stress Ulcer Prophylaxis as per unit policy

Prognostication and Communication.

Daily neurological examination, but do not prognosticate under effect of sedatives or TTM.

If poor neurological status, Request CT head at 48hrs if not done already. If no hypoxic ischemic encephalopathy evident on CT, request EEG **AND** Somato-Sensory Evoked Potentials if GCS ≤5 (both via EEG dept) and Neurone Specific Enolase at 48-72hrs post-ROSC (available via WCP).

Neuroprognostication and communication with families should be lead by an ICM consultant, using the ResusUK algorithm (Neuroprognostication Algorithm 2021.pdf (resus.org.uk)), see page 3.

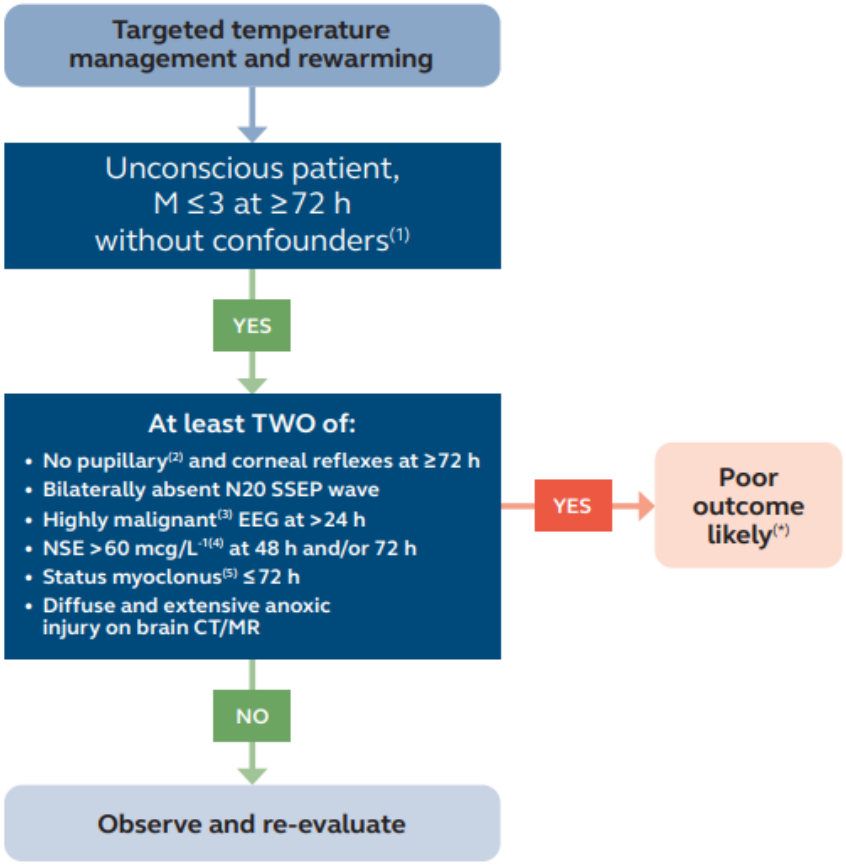
References

1) Post-resuscitation care Guidelines | Resuscitation Council UK
2) Neuroprognostication Algorithm 2021.pdf (resus.org.uk)

Resuscitation Council UK

GUIDELINES 2021

Neuroprognostication of the comatose adult patient after resuscitation from cardiac arrest



[1] Major confounders may include analgo-sedation, neuromuscular blockade, hypothermia, severe hypotension, hypoglycaemia, sepsis, and metabolic and respiratory derangements

[2] Use an automated pupillometer, when available, to assess pupillary light reflex

[3] Suppressed background ± periodic discharges or burst-suppression, according to American Clinical Neurophysiology Society

[4] Increasing NSE values between 24–48 h or 24/48 and 72 h further support a likely poor outcome

[5] Defined as a continuous and generalised myoclonus persisting for 30 min or more

[*] Caution in case of discordant signs indicating a potentially good outcome